

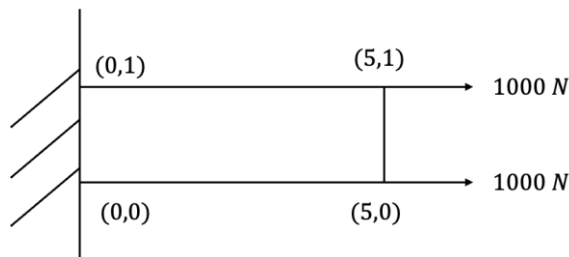
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**Intro to Finite Element Method (CIV E – 665)**  
Assignment No. 7

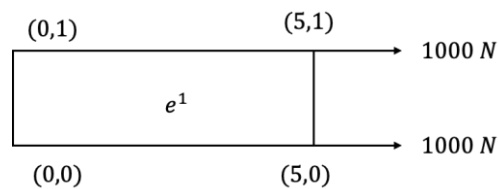


**Question 1 (50 Points)**

Consider the following structure. Solve for the nodal displacements assuming a plane stress condition (with  $t = 0.2$  m) and that the elastic modulus and Poisson's ratio are 100 GPa and 0.3 respectively. Assume that the structure has no body forces.



Solve for the **nodal displacements** using **one linear rectangular** element as shown below:



**NB: The coordinates are expressed in m.**

For 2D linear rectangular elements, the system matrix can be expressed as

$$\left[ \int_{\Omega} [B]^T [C] [B] t dx dy \right] \{u\} = \left\{ \int_{\Omega} [\psi]^T \left\{ \begin{matrix} f_x \\ f_y \end{matrix} \right\} t dx dy \right\} + \oint_{\Gamma} [\psi]^T \left\{ \begin{matrix} t_x \\ t_y \end{matrix} \right\} t dS \right\}$$

where:

$$B = \begin{bmatrix} \frac{\partial \psi_1}{\partial x} & 0 & \frac{\partial \psi_2}{\partial x} & 0 & \frac{\partial \psi_3}{\partial x} & 0 & \frac{\partial \psi_4}{\partial x} & 0 \\ 0 & \frac{\partial \psi_1}{\partial y} & 0 & \frac{\partial \psi_2}{\partial y} & 0 & \frac{\partial \psi_3}{\partial y} & 0 & \frac{\partial \psi_4}{\partial y} \\ \frac{\partial \psi_1}{\partial y} & \frac{\partial \psi_1}{\partial x} & \frac{\partial \psi_2}{\partial y} & \frac{\partial \psi_2}{\partial x} & \frac{\partial \psi_3}{\partial y} & \frac{\partial \psi_3}{\partial x} & \frac{\partial \psi_4}{\partial y} & \frac{\partial \psi_4}{\partial x} \end{bmatrix}$$

For this case, the numbering system is shown as the Figure 1.

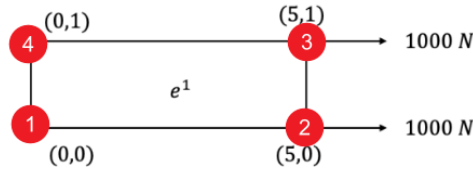


Figure 1: Numbering system

$$\psi_1 = \left(1 - \frac{\bar{x}}{a}\right) \left(1 - \frac{\bar{y}}{b}\right) \quad \psi_2 = \frac{\bar{x}}{a} \left(1 - \frac{\bar{y}}{b}\right) \quad \psi_3 = \frac{\bar{x}}{a} \frac{\bar{y}}{b} \quad \psi_4 = \left(1 - \frac{\bar{x}}{a}\right) \left(\frac{\bar{y}}{b}\right)$$

where a and b are the length and width, respectively, so  $a = 5$  and  $b = 1$  in this case.

## 0.1 B matrix

Substitute the given data in to  $B$  and  $\psi_i$

$$\psi_1 = \left(1 - \frac{\bar{x}}{5}\right) \left(1 - \frac{\bar{y}}{1}\right) \quad \psi_2 = \frac{\bar{x}}{5} \left(1 - \frac{\bar{y}}{1}\right) \quad \psi_3 = \frac{\bar{x}}{5} \frac{\bar{y}}{1} \quad \psi_4 = \left(1 - \frac{\bar{x}}{5}\right) \left(\frac{\bar{y}}{1}\right)$$

B =

$$\begin{bmatrix} y/5 - 1/5, & 0, & 1/5 - y/5, & 0, & y/5, & 0, & -y/5, & 0 \\ 0, & x/5 - 1, & 0, & -x/5, & 0, & x/5, & 0, & 1 - x/5 \\ x/5 - 1, & y/5 - 1/5, & -x/5, & 1/5 - y/5, & x/5, & y/5, & 1 - x/5, & -y/5 \end{bmatrix}$$

## 0.2 C matrix

Use matlab to get the  $[c]$ , for plane stress assumption:

$$[c] = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} = \frac{100 \times 10^9}{0.91} \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix}$$

### 0.3 K matrix

Substitute  $[c]$ ,  $t$  and  $[B]$  into Equation 1, thus we can get  $[Ke1]$

$$[Ke1] = \left[ \int_{\Omega} [B]^T [C] [B] t dx dy \right] \quad (1)$$

```
Ke1 =
[ 1.429e+10,  3.571e+9,  4.945e+9, -2.747e+8, -7.143e+9, -3.571e+9, -1.209e+10,  2.747e+8]
[  3.571e+9,  3.714e+10,  2.747e+8,  1.78e+10, -3.571e+9, -1.857e+10, -2.747e+8, -3.637e+10]
[  4.945e+9,  2.747e+8,  1.429e+10, -3.571e+9, -1.209e+10, -2.747e+8, -7.143e+9,  3.571e+9]
[ -2.747e+8,  1.78e+10, -3.571e+9,  3.714e+10,  2.747e+8, -3.637e+10,  3.571e+9, -1.857e+10]
[ -7.143e+9, -3.571e+9, -1.209e+10,  2.747e+8,  1.429e+10,  3.571e+9,  4.945e+9, -2.747e+8]
[ -3.571e+9, -1.857e+10, -2.747e+8, -3.637e+10,  3.571e+9,  3.714e+10,  2.747e+8,  1.78e+10]
[ -1.209e+10, -2.747e+8, -7.143e+9,  3.571e+9,  4.945e+9,  2.747e+8,  1.429e+10, -3.571e+9]
[  2.747e+8, -3.637e+10,  3.571e+9, -1.857e+10, -2.747e+8,  1.78e+10, -3.571e+9,  3.714e+10]
```

### 0.4 Element wise

Based on our numbering system and the given data.

Node 1: (0, 0) – Fixed Node 2: (5, 0) – Free, subjected to  $F_x = 1000$  N Node 3: (5, 1) – Free, subjected to  $F_x = 1000$  N Node 4: (0, 1) – Fixed

So, the displacements at node 1 and 4 are zero, that is:  $u_{1x} = u_{1y} = u_{4x} = u_{4y} = 0$

$$U = \begin{bmatrix} 0 \\ 0 \\ u_{2x} \\ v_{2y} \\ u_{3x} \\ v_{3y} \\ 0 \\ 0 \end{bmatrix}$$

Apply the boundary condition:

$$f = \begin{bmatrix} 0 \\ 0 \\ 1000 \\ 0 \\ 1000 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Use zero to replace the corresponding rows and columns and then we can use matlab to solve the Equation 2

$$[Ke1][U] = [f] \quad (2)$$

Displacements at free nodes:

| DOF     | Value_m     |
|---------|-------------|
| -----   | -----       |
| {'u2x'} | 4.8781e-07  |
| {'u2y'} | 2.1875e-08  |
| {'u3x'} | 4.8781e-07  |
| {'u3y'} | -2.1875e-08 |

The source code please see in the flowing.

```
clear; clc; close all;
%% Constants
E=100E9; mu=0.3; t=0.2;
% c11=E*(1-mu)/((1+mu)*(1-2*mu));
%Material constants for plane strain
% c22=c11; c12=E*mu/((1+mu)*(1-2*mu)); c66=E/(2*(1+mu));
%Material constants for plane stress
c11 =1; c12 =mu
c22=c11; c21=c12; c66 = (1-mu)/2;

C=E/(1-mu^2)*[c11 c12 0;c12 c22 0;0 0 c66]

%% Shape functions
syms x y
psi11=(1-x/5)*(1-y);
psi21=x/5*(1-y);
psi31=x*y/5;
psi41=(1-x/5)*y;
%Shape Function Matrix
psi_e1=[psi11 0 psi21 0 psi31 0 psi41 0;
        0 psi11 0 psi21 0 psi31 0 psi41];

%Strain Displacement Matrix
z = sym(0);

Be1=[diff(psi11,x) z diff(psi21,x) z diff(psi31,x) z diff(psi41,x) z;
     z diff(psi11,y) z diff(psi21,y) z diff(psi31,y) z diff(psi41,y);
     diff(psi11,y) diff(psi11,x) diff(psi21,y) diff(psi21,x) diff(psi31,y) diff(psi31,x) diff(psi41,y) diff(psi41,x)];

Ke1=t*int(int(transpose(Be1)*C*Be1,y,[0,1]),x,[0,5]);
disp(vpa(Ke1,4))

syms u2x u2y u3x u3y;

U = [0;0;u2x;u2y;u3x;u3y;0;0]
F = [0;0;1000;0;1000;0;0;0]

free_dofs = [3,4,5,6];
K_reduced = Ke1(free_dofs, free_dofs);
```

```

F_reduced = F(free_dofs);

% Solve the system: K_reduced * U_free = F_reduced
U_free = K_reduced \ F_reduced;

% Assign the solved values back to the full displacement vector
U(free_dofs) = U_free;

%% Display Results
fprintf('Displacements at free nodes:\n');
disp(table({'u2x'; 'u2y'; 'u3x'; 'u3y'}, double(U_free), ...
    'VariableNames', {'DOF', 'Value_m'}))

```

**Question 2 (20 Points)**

Evaluate the Jacobian matrix for a quadrilateral element bounded by (0,0), (2,2), (2,1) and (-1,0.5)

**Question 3 (30 Points)**

Solve the deep beam tutorial ([Problem 1: Deep Beam](#)) provided from Dr. Adeeb's course textbook on ABAQUS. To show the completion of this exercise show a screenshot of  $S_{11}$

## 1 Solution for question 2

The element is shown as the Figure 2.

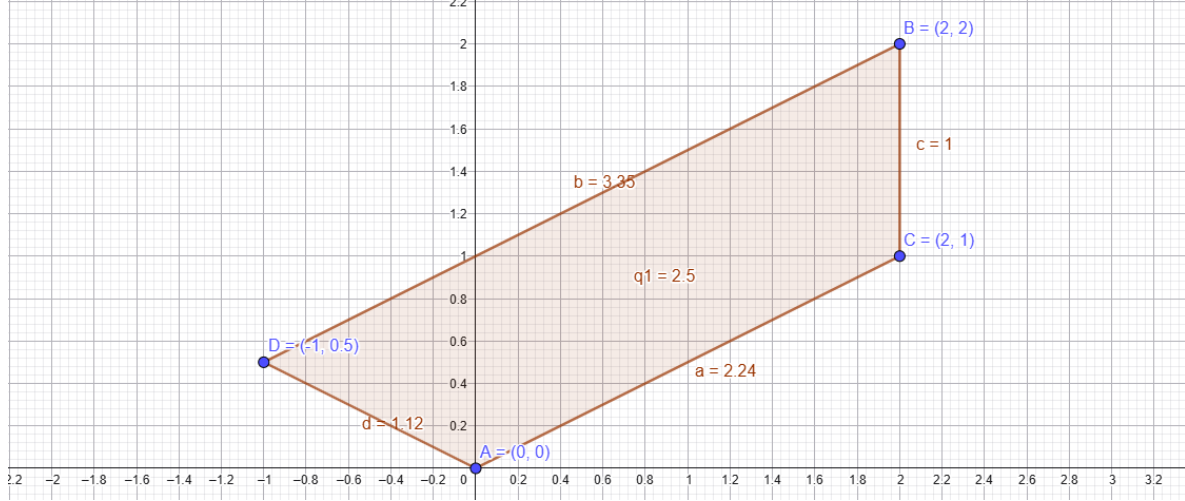


Figure 2: The geometray shape for the given data

We should re-number this element by [A,C,B,D], which maps to [Node1, Node2, Node3, Node4].

For an isoparametric quadrilateral element, we can use the relationships we found between  $(x,y)$  and  $(\zeta,\eta)$

$$\begin{aligned}
 x &= \sum_{j=1}^4 x_j N_j = x_1 N_1 + x_2 N_2 + x_3 N_3 + x_4 N_4 \\
 &= 2N_2 + 2N_3 - 1N_4 \\
 y &= \sum_{j=1}^4 y_j N_j = y_1 N_1 + y_2 N_2 + y_3 N_3 + y_4 N_4 \\
 &= 1N_2 + 2N_3 + 0.5N_4
 \end{aligned}$$

And the shape function:

$$N_1 = \frac{(1-\zeta)(1-\eta)}{4} \quad N_2 = \frac{(1+\zeta)(1-\eta)}{4} \quad N_3 = \frac{(1+\zeta)(1+\eta)}{4} \quad N_4 = \frac{(1-\zeta)(1+\eta)}{4}$$

The Jacobian matrix is:

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$

Then we substitute the each node coordinate in to  $[J]$ , About  $J_{11}$

$$J_{11} = \frac{\partial x}{\partial \xi} = \sum_{i=1}^4 \frac{\partial N_i}{\partial \xi} x_i = \frac{1}{4} [-(1-\eta)(0) + (1-\eta)(2) + (1+\eta)(2) - (1+\eta)(-1)] = \frac{5+\eta}{4}$$

About  $J_{12}$

$$J_{12} = \frac{\partial y}{\partial \xi} = \sum_{i=1}^4 \frac{\partial N_i}{\partial \xi} y_i = \frac{1}{4} [-(1-\eta)(0) + (1-\eta)(1) + (1+\eta)(2) - (1+\eta)(0.5)] = \frac{\mathbf{2.5} + \mathbf{0.5}}{\mathbf{4}}$$

About  $J_{21}$

$$J_{21} = \frac{\partial x}{\partial \eta} = \sum_{i=1}^4 \frac{\partial N_i}{\partial \eta} x_i = \frac{1}{4} [-(1-\xi)(0) - (1+\xi)(2) + (1+\xi)(2) + (1-\xi)(-1)] = \frac{-\mathbf{1} +}{\mathbf{4}}$$

About  $J_{22}$

$$J_{22} = \frac{\partial y}{\partial \eta} = \sum_{i=1}^4 \frac{\partial N_i}{\partial \eta} y_i = \frac{1}{4} [-(1-\xi)(0) - (1+\xi)(1) + (1+\xi)(2) + (1-\xi)(0.5)] = \frac{\mathbf{1.5} + \mathbf{0.5}}{\mathbf{4}}$$

Thus,

$$[J] = \begin{bmatrix} \frac{5+\eta}{4} & \frac{2.5+0.5\eta}{4} \\ \frac{-1+\xi}{4} & \frac{1.5+0.5\xi}{4} \end{bmatrix}$$

## 2 Solution for question 3

I simulated the full scale geometry, so it's longer than the model in the tutorial video, as shown in th Figure 3. There is a little bit of difference between my results and the results of video. I thought its from the constraint setting between reference point and the top surface coupling.

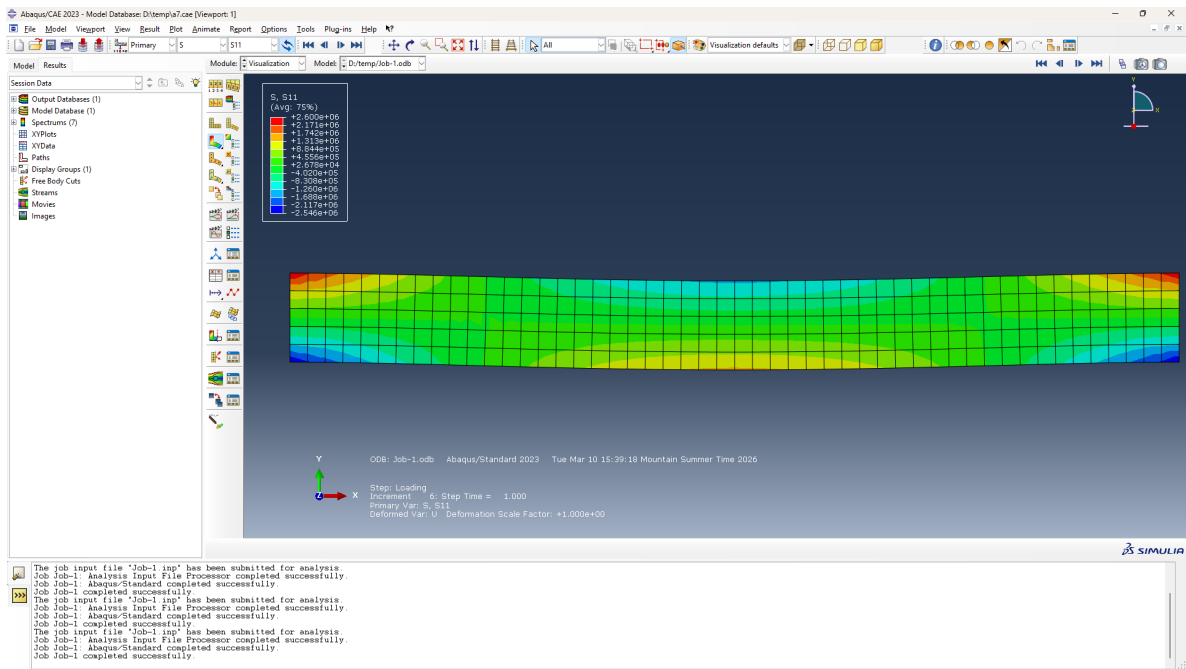


Figure 3: The S11 of the problem